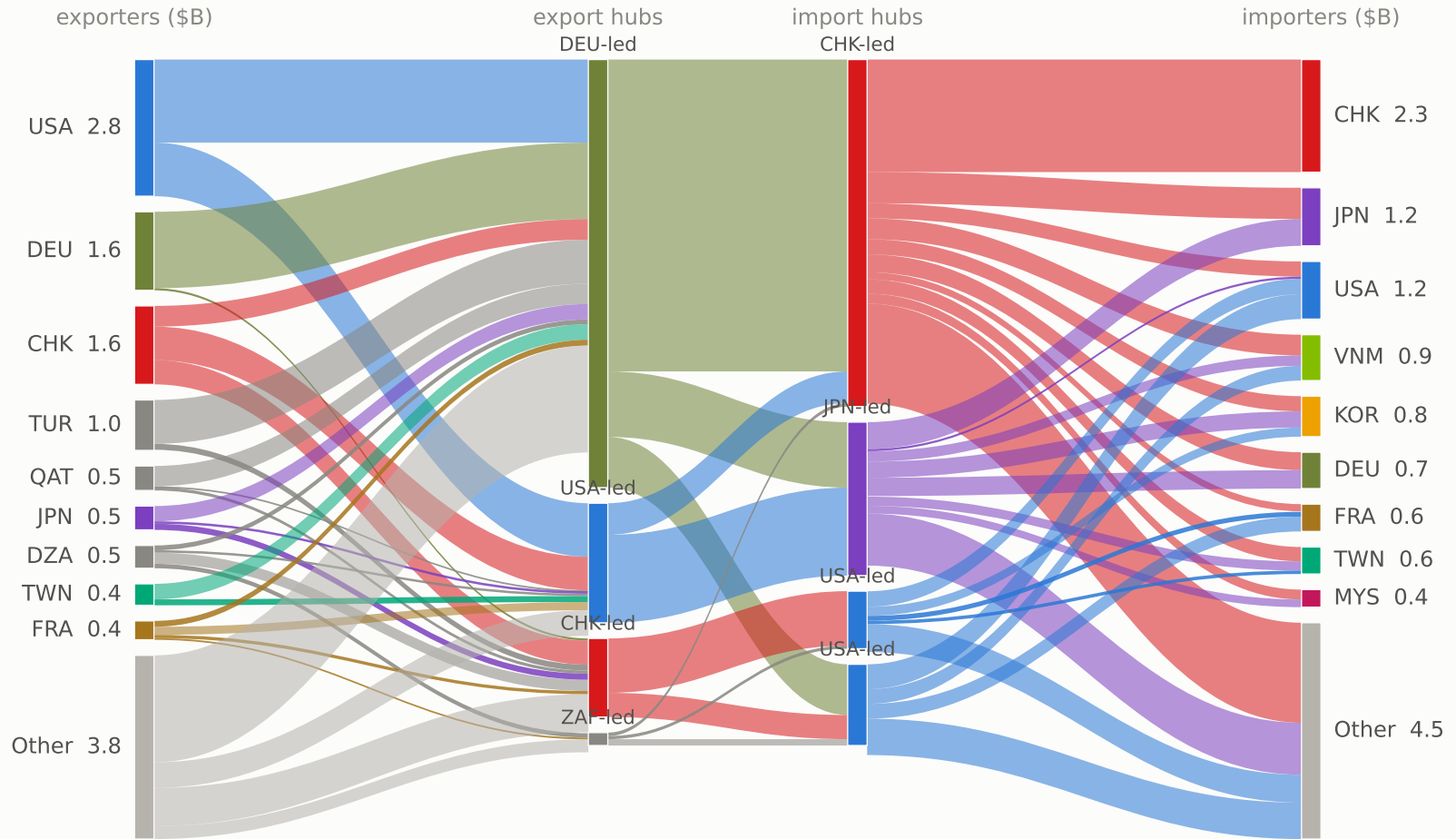


Stage 1 — Raw materials [unrotated] (silicon 280461, rare gases, Ga/Ge, chemicals) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$13B



Hub decomposition from the factor model (UNROTATED spectral basis, hubs in descending eigenvalue order; constant-loading annual MFM 2020-24 on Atlas bilateral data); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral, all countries.